Technical Specifications and Engineering Data

Project: **Adaptive PLC-Based Control System for the Gate Valve of a 10 kW Micro Pumped-Storage Hydropower Plant.**

Table 1. Comparison of the main motor types for small valves in micro-PSH.

Motor Type	Voltage	Advantages	Disadvantages	Application for micro-PSH
Asynchronous (AC)	220–380 V	Simple, reliable, and durable	More difficult to control speed without a frequency converter	Stationary valves with fixed speed
Brushed (DC)	12–220 V	Easy to control speed and direction	Brushes require maintenance, shorter service life	Compact low-power drives with precise positioning
Brushless DC (BLDC)	12–220 V	Brushless, high reliability, smooth operation	Requires an electronic driver	Modern small drives with energy efficiency
Servo motor	12–220 V, AC	Precise positioning, high dynamic performance	Expensive, often excessive for a simple valve	Automation where position control is required
Worm gear motor	AC or DC	Compact, high torque, self-locking	Lower efficiency (≈40–60 %)	Most common for small valves with diameters of 100–150 mm

Table 2. Photometric Sensors.

Model / Manufacturer	Sensor Type / Application	Range, Lux	Country / Brand	Notes
FST100-2003	Industrial light sensor with 4–20 mA analog output	0–200 000 lx	China	Optimal for PLCs and industrial automation
BH1750	Digital photometric light sensor with digital output (I^2C)	1–65 000 lx	China / Japan	Compact and precise, suitable for microcontrollers; not suitable for PLCs
SE-100 Photometric Sensor	Industrial photometric sensor with 0–2000 mV analog output	0-150 000 lx	APOGEE (USA)	Can be integrated into automation, but requires signal conversion
LDR Photodetector Module (China)	Photosensor for control with resistance/voltage change output	10-100 000 lx	China	Suitable for simple on/off control; low stability

Table 3. Recommendations for PLC types.

Category	Advantages	Notes
Siemens S7-1200, AI 4×U/I module (part number 6ES7231-4HD32-0XB0)	-Fully industrial standard -Built-in 4–20 mA analog inputs -Easy SCADA integration -Expandable with network capability	Higher cost than budget models
Delta DVP ES2 / Delta DVP SX	-Low cost, compact -Simple setup for a single sensor -2–4 analog inputs	-Limited expansion capabilities -Requires an external interface for SCADA
WAGO 750 8x0	-Modular system, easy to add new sensors -Support for network interfaces (Modbus,EtherCAT) -High industrial reliability	-More complex setup -Price higher than budget options